(Please write your Enrollment Number) Enrollment No._____

END-TERM EXAMINATION

(CBCS)

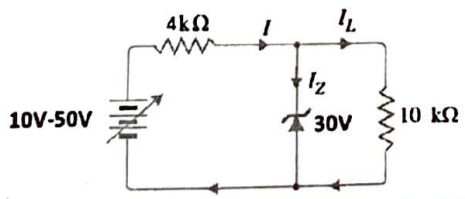
B. Tech (IT/MAE/DMAM), 2nd Semester

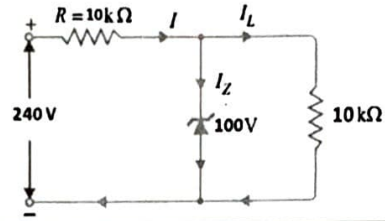
(August, 2022)

Subject Code: BAS-104	Subject: Applied Physics - II

Time : 3 Hours **Maximum Marks : 60**

Note: Q. 1 is compulsory. Attempt any one question from each unit.

Q1	Do any four from the following	**(4*5)**
	(a) What is skin depth? Suppose you wish to design a microwave experiment to operate at a frequency of 10 MHz. How thick would you like to make the silver coating so that the microwaves do not penetrate the coating? The conductivity of silver is $6.2 \times 10^7\ S/m$ and its relative permeability≈ 1.	
	(b) Draw the planes for which the Miller indices are (112), (200), (120), (210) and (221)	
	(c) A parallel plate capacitor has square plates of side 5 cm and separated by a distance of 1 mm. (a) Calculate the capacitance of this capacitor. (b) If a 10 V battery is connected to the capacitor, what is the charge stored in any one of the plates? (The value of $\varepsilon_o = 8.85 \times 10^{-12}\ Nm^2\ C^{-2}$)	
	(d) What is Poynting's Theorem? Discuss physical significance of Poynting vector.	
	(e) What is selectivity and sensitivity of a sensor? Why it is important in the case of chemical sensors? Explain.	
	Unit 1	
Q2	(a) Electromagnetic waves are traveling in dielectric medium, using Maxwell's equations find the wave equations for electric field and magnetic field of EM wave and also find the speed of EM wave. The speed of the EM wave will be less or more in dielectric medium as compare to free space?	**(5,5)**
	(b) Write all Maxwell's equations in differential form and discuss their physical significance.	
Q3	(a) Derive Maxwell's 1st and 3rd equation and discuss their physical significance.	**(5,5)**
	(b) What are electromagnetic waves? Show that the electromagnetic waves are transverse in nature.	
	Unit 2	
Q4	(a) What is the difference between intrinsic and extrinsic semiconductors? Draw the fermi energy level diagram of the doped and undoped semiconductor.	**(5,5)**
	(b) For the following circuit (i), find the maximum and minimum values of Zener diode current 	

Q5	(a) Show that the fermi level of n-type semiconductor lies near conduction band also show it with the help of diagram	(5,5)
	(b) For the circuit shown below (i), find (i) the output voltage (ii) the voltage drops across series resistance (iii) the current through Zener diode. 	
	Unit 3	
Q6	(a) What do you mean by the resolution of a sensor? There are three sensors A, B, C, out of them sensor A response for the minimum change of input voltage of order 0.4 V, while the other sensor B, C response for 0.1 V and 0.6 V input voltage change which of them you will choose for higher resolution and why?	(5,5)
	(b) What is sensor? Does the sensor and the transducer are the same thing, explain. What is transfer function of a sensor?	
Q7	(a) What is the difference between direct and complex sensor? What do you understand by the term sensitivity and non-linearity of a sensor?	(5,5)
	(b) For a sensor electrical resistance changes when the material is mechanically deformed. The transfer function for a sensor is given by $R = \frac{\rho l^2}{V}$. Find the sensitivity of the sensor. If we use two wire one of them is 5 cm long and diameter of cross section is 3mm, and other is 2 cm long and having diameter of cross section is 6 mm, which wire will be highly sensitive and why?	
	Unit 4	
Q8	(a) What is the advantage of SEM over normal microscope? Draw the diagram of SEM and discus its working principle and all components of SEM.	(5,5)
	(b) What is CT scan? Discuss its working principle and how it is different form normal X-RAY technique.	
Q9	(a) What are the different phenomena that occurs when an electron beam strike on the surface of the sample in the case of SEM? What are the different applications of SEM.	(5,5)
	(b) What are the advantages and disadvantages of CT scan? What is the working principle of SEM? Discuss.	